LECTURE 00 / INTRODUCTION

Cryptography

Trust in a connected world



EVERYDAY LIFE

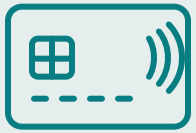
You used it before class.

Which of these have you done today?



Open a website

WEB



Tap to pay

PAYMENTS



Sign in

IDENTITY



Install an update

SOFTWARE

Three questions behind digital trust.

Cryptography uses mathematics to protect information.

Who can read it?

Confidentiality



Keep the message private.

Has it changed?

Integrity



Detect an altered message.

Who is on the other end?

Authentication



Check the claimed identity.

Cryptanalysis asks how these protections could fail.

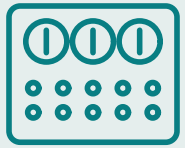
New tools. New ways to communicate.

As communication and computing changed, the methods of protection changed too.



Pen and paper

Letters and couriers



Machines

Military radio traffic



Connected devices

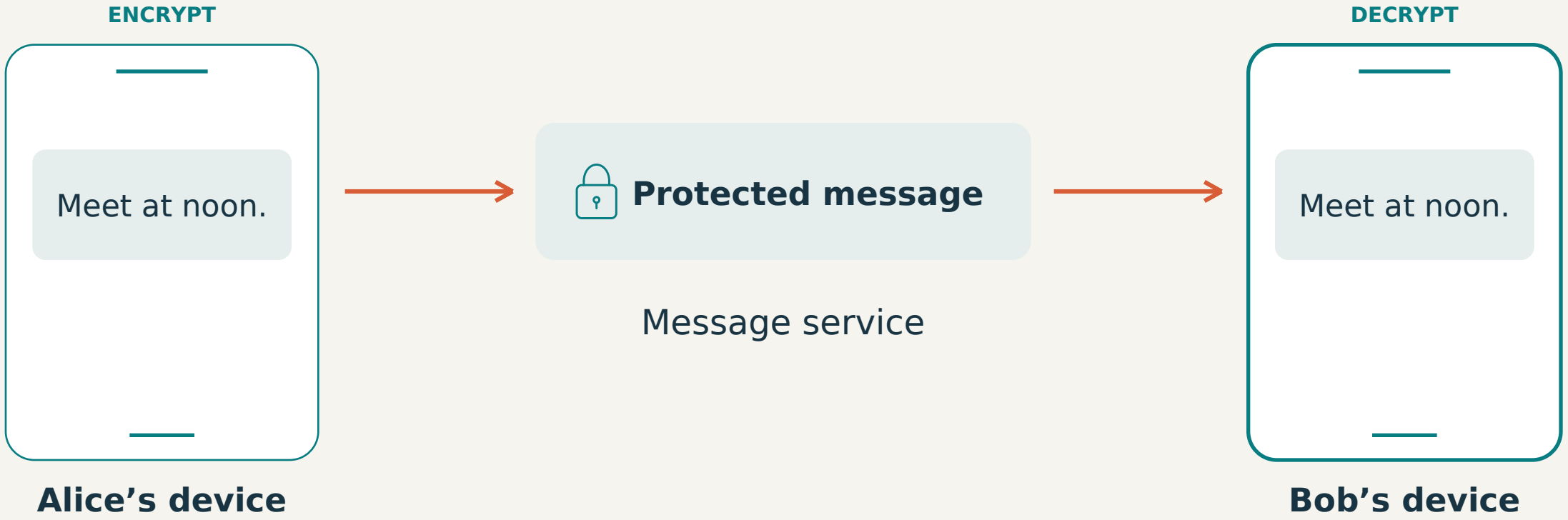
Global digital communication

Enigma is one historical example; its cryptanalysis was a collective international effort.

MESSAGING

Private words. Public networks.

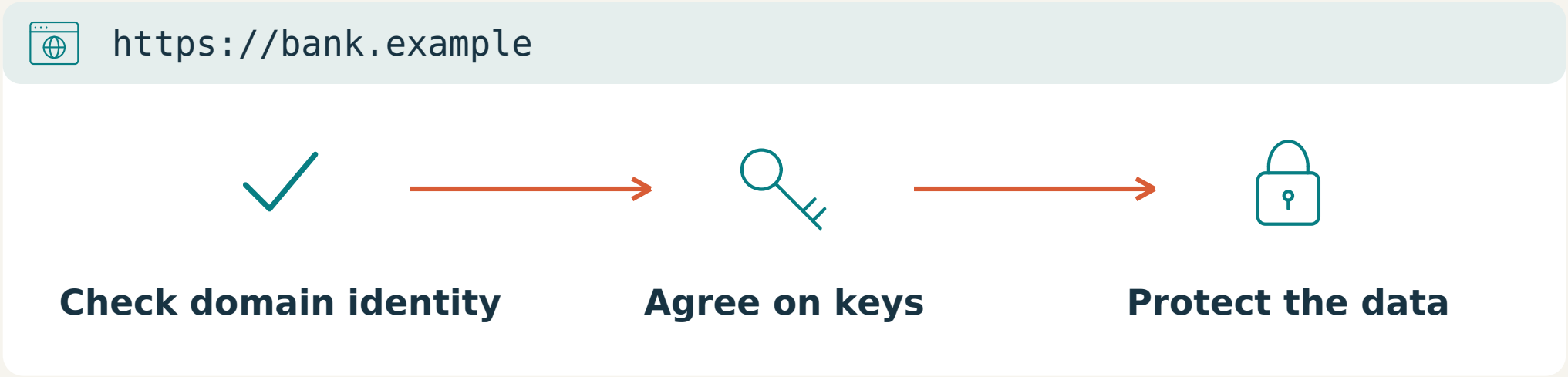
End-to-end encryption protects a conversation’s content as it crosses the network.



Keys let the intended devices unlock the message.

How do strangers trust a connection?

Your browser checks the server’s identity for the named domain.



Two computers can create a protected connection without sharing a secret in advance.

One tap. A chain of trust.

Modern payments use cryptographic checks to authenticate transaction data.



Card or phone

Payment terminal

Bank

Protect payment data

Authenticate the transaction

Detect changes

SIGNING IN

Your face can unlock a key.

Your device proves it can use the right key.



The website receives cryptographic proof, not your fingerprint or face scan.

DOCUMENTS AND SOFTWARE

Trust travels with the data.

Digital signatures help check origin and detect changes.



Electronic passport

Check the issuer and integrity of signed chip data.



Software update

Check the publisher and integrity of an update.

Authenticity matters even when the information itself is not secret.

Cryptography is part of the system.

Strong cryptography cannot make every decision safe.

ILLUSTRATIVE SCAM

“Please approve this transfer to keep your account safe.”



Cryptography



Devices & software



People & processes

A payment can pass the checks and still be part of a scam.

Next: ciphers you can see inside.

We begin with historical examples that we can work through by hand.

MEET → PHHW

How do we hide a message?

How could someone recover it?